

Korean SMOG Model Coefficient and Fit Analysis

Model output review

16 September 2026
Saved model results through 2026Q1

1 Scope and interpretation

This report analyzes the standalone Korean Python SMOG model. Estimates come from its saved parameter JSON and accepted-iteration likelihood history. The generation sample is 2000Q1–2022Q4 (92 quarters). The update applies those fixed coefficients to later data; the latest complete saved result is 2026Q1. The latent potential output and NAIRU are model estimates, not directly observed quantities.

2 Results and charts

The fixed-parameter update runs through 2026Q1. Its saved smoothed output gap is -0.052% of potential output and its NAIRU is 2.761% . Observed unemployment is 2.867% , or 0.106 percentage points above the fitted NAIRU. Smoothed states use the full updated sample; filtered states use observations available through each plotted quarter, with the same historical coefficients. The charts are model estimates, not direct measurements of potential output or the NAIRU.

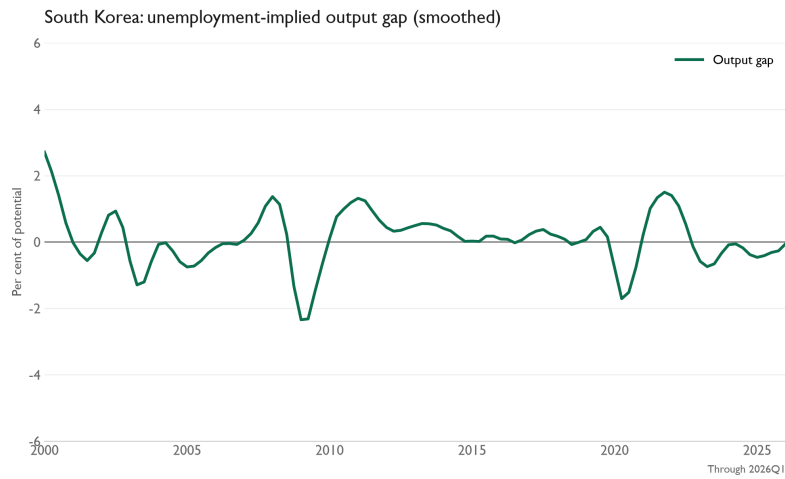


Figure 1: Smoothed output gap.

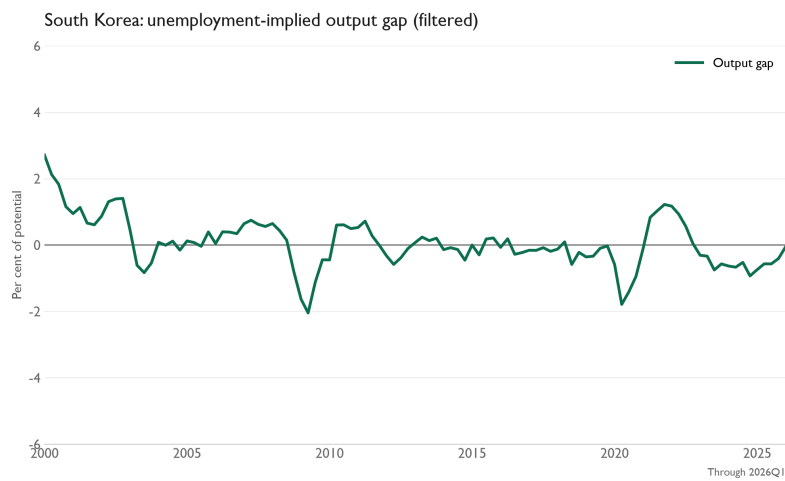


Figure 2: Filtered output gap.

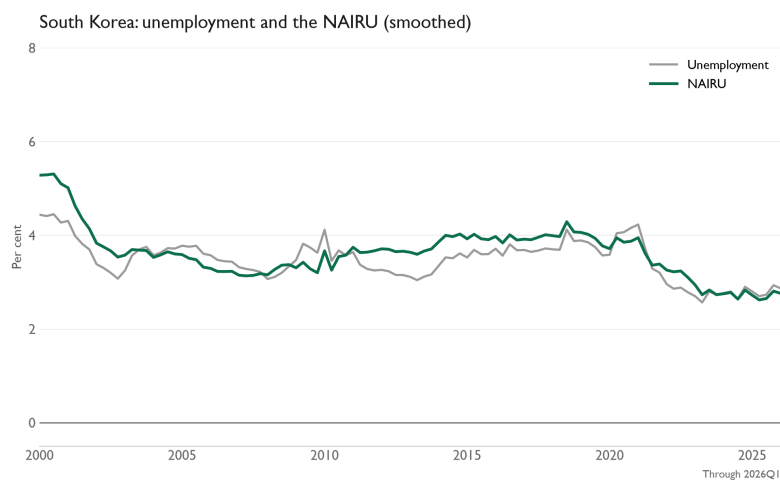


Figure 3: Observed unemployment and smoothed NAIRU.

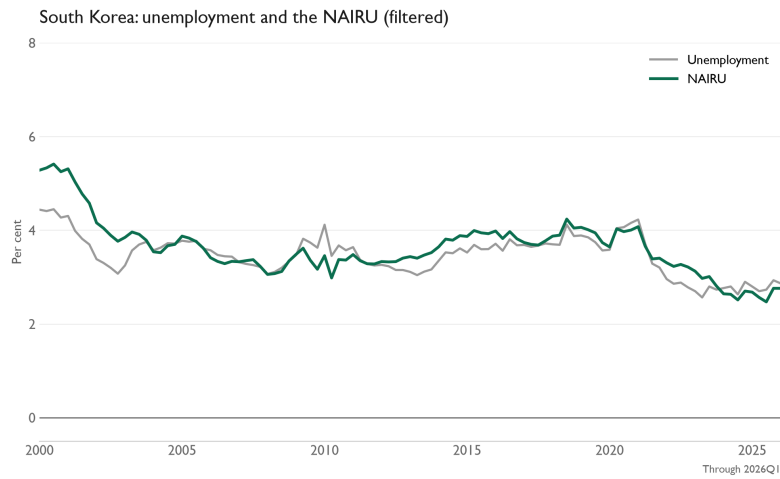


Figure 4: Observed unemployment and filtered NAIRU.

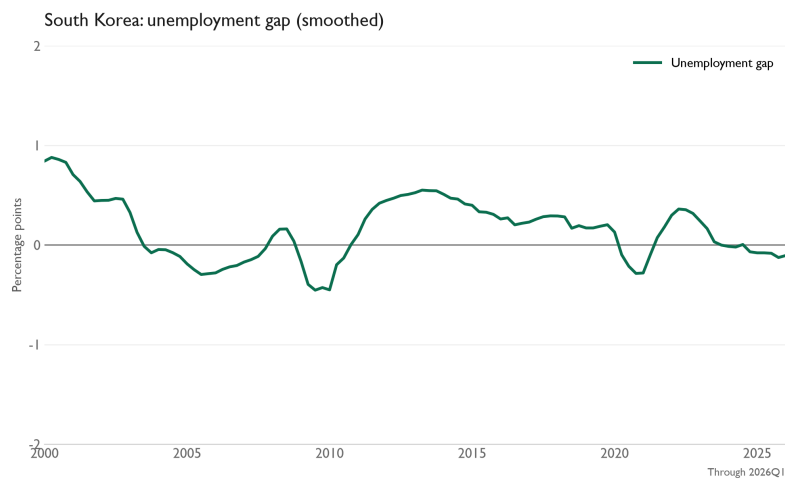


Figure 5: Smoothed NAIRU minus observed unemployment.

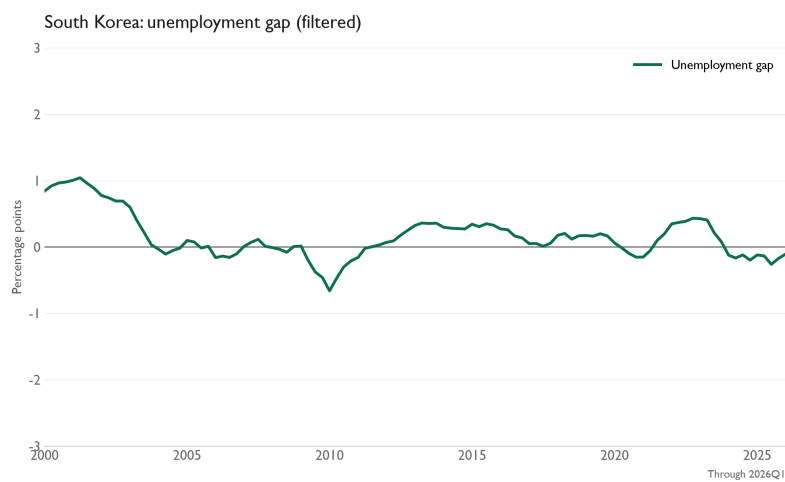


Figure 6: Filtered NAIRU minus observed unemployment.

3 Reading the current output gap

The latest saved Korean gap is -0.052% in 2026Q1, quite close to zero; observed unemployment is 0.106 percentage points above the fitted NAIRU. The local core-inflation reading is 2.176% and the expectation input is 2.633%. The smoothed gap contributes just -0.006 percentage points to the fitted inflation equation. Its full conditional fitted value is 2.153%, leaving an actual-minus-fitted difference of 0.022 points. This uses the smoothed state and is not an out-of-sample forecast.

The evidence here is weak for calling the output gap materially too negative: its estimated level and direct inflation contribution are both small. The unemployment equation helps locate the latent gap, but lagged inflation, expectations, import prices, the wage signal, and measurement noise also shape the joint fit. Current inflation alone cannot identify which latent state or coefficient should change.

4 State-space structure

The model uses seven states, $s_t = (g_t, g_{t-1}, g_{t-2}, y_t^*, y_{t-1}^*, u_t^*, u_{t-1}^*)'$, where y_t is log output, $g_t = y_t - y_t^*$ is a log output gap, and u_t^* is the latent structural unemployment rate. The reported output gap is $100g_t$, in percent of potential output. The transition equations are

$$g_t = \phi_1 g_{t-1} + \phi_2 g_{t-2} + \eta_t^g, \quad (1)$$

$$y_t^* = y_{t-1}^* + g^* + \eta_t^y, \quad (2)$$

$$u_t^* = u_{t-1}^* + \eta_t^u. \quad (3)$$

The output gap follows an AR(2), potential log output is a random walk with quarterly drift g^* , and the NAIRU is a random walk. The three state shocks are independent in the implemented diagonal covariance matrix. The state vector also carries lags needed by the observation equations.

At the beginning of each generation run, HP-based gap, potential output, and NAIRU values set a known initial state with zero prior covariance. The model reads its historical HP columns from the Generation tab. The HP series initialize the filter and provide regression-based optimizer starting values; the saved maximum-likelihood coefficients are fitted by Python from the country's own signals. This fixed initialization can matter for early likelihood terms and the resulting parameter estimates.

5 Korea: equations and estimates

Korea has four observed signals: log GDP y_t , unemployment u_t , core inflation π_t , and nominal unit labour cost growth w_t . Here π_t^e is the Bank of Korea expectation input and p_{t-1}^m is the lagged four-quarter import-price change. The implemented observation equations are

$$y_t = y_t^* + g_t + \varepsilon_t^y, \quad (4)$$

$$u_t = u_t^* + \lambda_3 g_t + \rho_1(u_{t-1} - u_{t-1}^*) + \varepsilon_t^u, \quad (5)$$

$$\pi_t = (1 - \beta_1 - \beta_2)\pi_t^e + \beta_1\pi_{t-1} + \beta_2\pi_{t-2} + \lambda_1 g_t + \psi p_{t-1}^m + \varepsilon_t^\pi, \quad (6)$$

$$w_t = (1 - \beta_4)\pi_t^e + \beta_4\pi_{t-1} + \lambda_2(u_{t-1} - u_{t-1}^*) + \varepsilon_t^w. \quad (7)$$

The wage equation connects labour-market slack to unit labour cost growth. Because the gap is in log units, a one percentage point change in the output gap multiplies λ_1 or λ_3 by 0.01.

Table 1: Korean maximum-likelihood coefficients. Values are rounded.

Parameter	Estimate	Meaning in the implemented model
β_1	1.311422	Weight on one-quarter lagged core inflation in the inflation equation.
β_2	-0.330134	Weight on two-quarter lagged core inflation.
β_4	1.966103	Weight on lagged inflation in the wage equation; the implied expectation weight is -0.9661 .
λ_1	11.745785	Inflation loading on the log output gap; a one percentage point gap gives about 0.1175 percentage points through this term.
λ_2	-8.279811	Wage loading on lagged unemployment minus lagged NAIRU; a one percentage point larger unemployment gap lowers predicted wage growth by about 8.28 percentage points, other terms fixed.
λ_3	-8.620103	Unemployment loading on the log output gap; a one percentage point output gap gives about -0.0862 percentage points.
ρ_1	0.889138	Persistence of lagged unemployment relative to lagged NAIRU.
ϕ_1	1.290464	First autoregressive coefficient of the latent output gap.
ϕ_2	-0.596930	Second autoregressive coefficient of the latent output gap.
ψ	0.002122	Loading of lagged four-quarter import-price change in inflation.
g^*	0.009052	Quarterly drift of potential log output, about 3.62% at an annualized log-growth rate.
ε_1	0.005770	Standard deviation of log GDP measurement noise; about 0.577% in log-output terms.
ε_2	0.074402	Standard deviation of unemployment measurement noise, in percentage points.
ε_3	0.285165	Standard deviation of core-inflation measurement noise, in percentage points.
ε_4	4.895773	Standard deviation of unit labour cost growth measurement noise, in percentage points.
ε_5	0.043368	Code-scale parameter for the gap shock: actual shock standard deviation is $\varepsilon_5/10 = 0.004337$ log units, or about 0.434%.
ε_6	0.006277	Standard deviation of the potential-log-output state shock.
ε_7	0.157126	Standard deviation of the quarterly NAIRU state shock, in unemployment percentage points.

The Korean inflation expectation weight is $1 - \beta_1 - \beta_2 = 0.0187$. The large wage measurement standard deviation, ε_4 , means that the fourth signal is comparatively noisy in its own units. It contributes to the likelihood, but an individual wage observation need not be close to its conditional fitted value. The sizeable negative λ_2 is a partial effect on the wage equation, conditional on its lagged inflation and expectation terms; it should not be read as a standalone causal estimate.

6 Likelihood fit and convergence

The script uses SciPy’s bounded L-BFGS-B optimizer on the Kalman-filter log likelihood. Starting guesses come from Korean historical data. The accepted-iteration history stores all 18 pa-

rameters, the full-sample log likelihood, and quarter-level contributions. Trial AR(2) coefficients outside the implemented stability inequalities are penalized.

Fit statistic	Korea
Historical quarters	92
Accepted iterations	129
Initial log likelihood	−33,775.3474
Iteration 25	12.1332
Iteration 50	12.3798
Final log likelihood	12.3840

The initial likelihood is extremely poor, but accepted steps increased it monotonically. The last history row agrees with the saved parameter JSON. Numerical convergence from these starting values and bounds does not establish a global optimum, tight identification, or forecast accuracy. The script does not export standard errors or out-of-sample forecast scores. Quarter-level likelihood contributions are Kalman one-step-ahead predictive density terms, not squared-error metrics. Early missing regressors cause the affected signals to be masked.

7 Reproduction and source files

From this model folder, run the generation script to re-estimate historical coefficients and overwrite the history CSVs. Then run the update script to apply the fixed fit to later quarters. Source files used for this report:

- Generation and update scripts in the **code** folder.
- Fitted parameter JSON at the model-folder root.
- The fit-history and likelihood-by-quarter CSVs in **output**.
- The updated state CSV in **results** and the **Update** tab of the local model-input workbook.

Numeric tables are a snapshot and do not update automatically when Python is rerun.